Project Name: “Arduino Uno Based Sun Tracking Solar System”.

Abstract:

The **"Arduino Uno Based Sun Tracking Solar System"** presents an innovative approach to enhance solar energy harvesting efficiency by implementing a sun-tracking mechanism using the Arduino Uno microcontroller. This system aims to optimize solar panel orientation, ensuring they continuously face the sun for maximum exposure, thus increasing energy yield. The project integrates light sensors and servo motors to detect sunlight direction and control panel positioning. The Arduino Uno's computational power enables real-time data processing and precise motor control. The proposed system offers a cost-effective solution for solar energy generation, particularly in locations with varying sunlight angles. Through its adaptable design and efficient tracking capabilities, this project contributes to advancing sustainable energy solutions and showcases the potential of combining IoT technology with renewable energy systems.

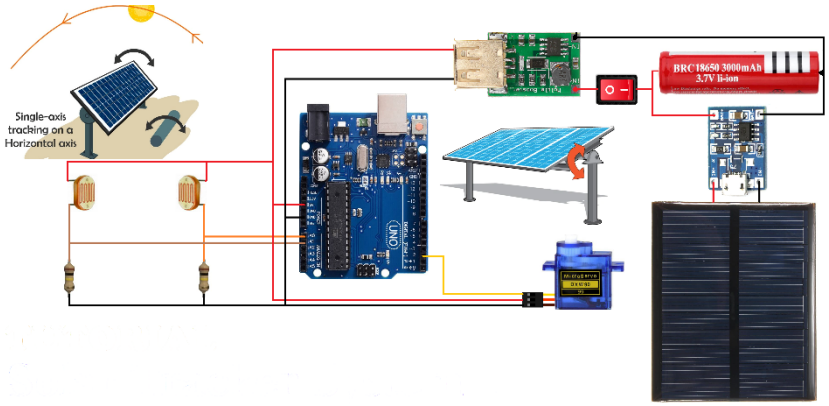

Fig 1: Structure of Sun Tracking Solar System.

Introduction:

Harnessing the power of the sun has become an important step towards efficient and sustainable energy generation in the constantly changing landscape of renewable energy options. The "Arduino Uno-Based Sun Tracking Solar System" stands out as a brilliant example of inventive technology as a result of the effort to maximize the exposure of solar panels to sunlight. High-efficiency solar panels are becoming increasingly necessary as the world's need for renewable energy rises. Inefficient energy conversion occurs when sunlight is captured by conventional stationary solar panels at fixed angles, especially when the angle of the sun changes during the day. This is where the Arduino Uno-based sun tracking system comes in. It is a state-of-the-art technology that dynamically aligns solar panels with the sun's trajectory to optimize their efficiency. This ground-breaking device relies mostly on the Arduino Uno microcontroller's abilities to direct the exact motions of solar panels. With the use of complex algorithms and real-time sensor inputs, the microcontroller coordinates the automatic adjustment of the solar panels to keep them pointed in the direction of the sun. Solar panels are more efficient even in areas with variable weather because to their dynamic tracking, which increases energy output while also extending the window during which they may be used. The convergence of cutting-edge electronics, computer programming, and renewable energy production is the focus of this project. In this investigation, we will examine the technical aspects of the Arduino Uno-based sun tracking system, including the choice of sensors, the programming of control algorithms, and the operation of the motorized panel. We will also go over the system's possible uses, which might include

everything from home solar setups to larger-scale installations, to show how it might alter the renewable energy landscape. Discover the complexities, advantages, and almost endless potential of the Arduino Uno-Based Sun Tracking Solar System as we set out on a voyage of creativity and ecological responsibility. This system illustrates our dedication to a greener, more sustainable future by fusing technology with environmental awareness.

Equipment (Parts & Tools):

1.	5mm Solid Wood Panel (60cm x 60 cm).	→1 pcs
2.	Hard Board Paper.	------
3.	Arduino Uno R3 (Italy).	→1 pcs
4.	Solar Panel 5V 1 Watt	→3 pcs
5.	LDR Sensor.	→2 pcs
6.	Mini Servo Motor SG90.	→1 pcs
7.	USB DC 1~5 V to DC 5V Voltage Step Up Boost Module.	→1 pcs
8.	LiPo Battery Charger Module Mini TP4056 IC.	→1 pcs
9.	Power On/Off Switch.	→1 pcs
10.	Battery Holder (Model: 18650).	→1 pcs
11.	Battery Cell 3.7V (Model: 18650).	→1 pcs
12.	Double Sided Form Tape.	→1 pcs
13.	Wire.	→3 feet
14.	Resistor 100k	→2 pcs
15.	Jumper Wire.	→3 pcs
16.	Soldering Iron.	→1 pcs
17.	Lead Wire.	→1 pcs
18.	Rosin.	→1 pcs
19.	Glue gun.	→1 pcs
20.	Glue sticks.	→2 pcs
21.	Wood Cutter.	→1 pcs
22.	Wire Cutter.	→1 pcs

Circuit Diagram:

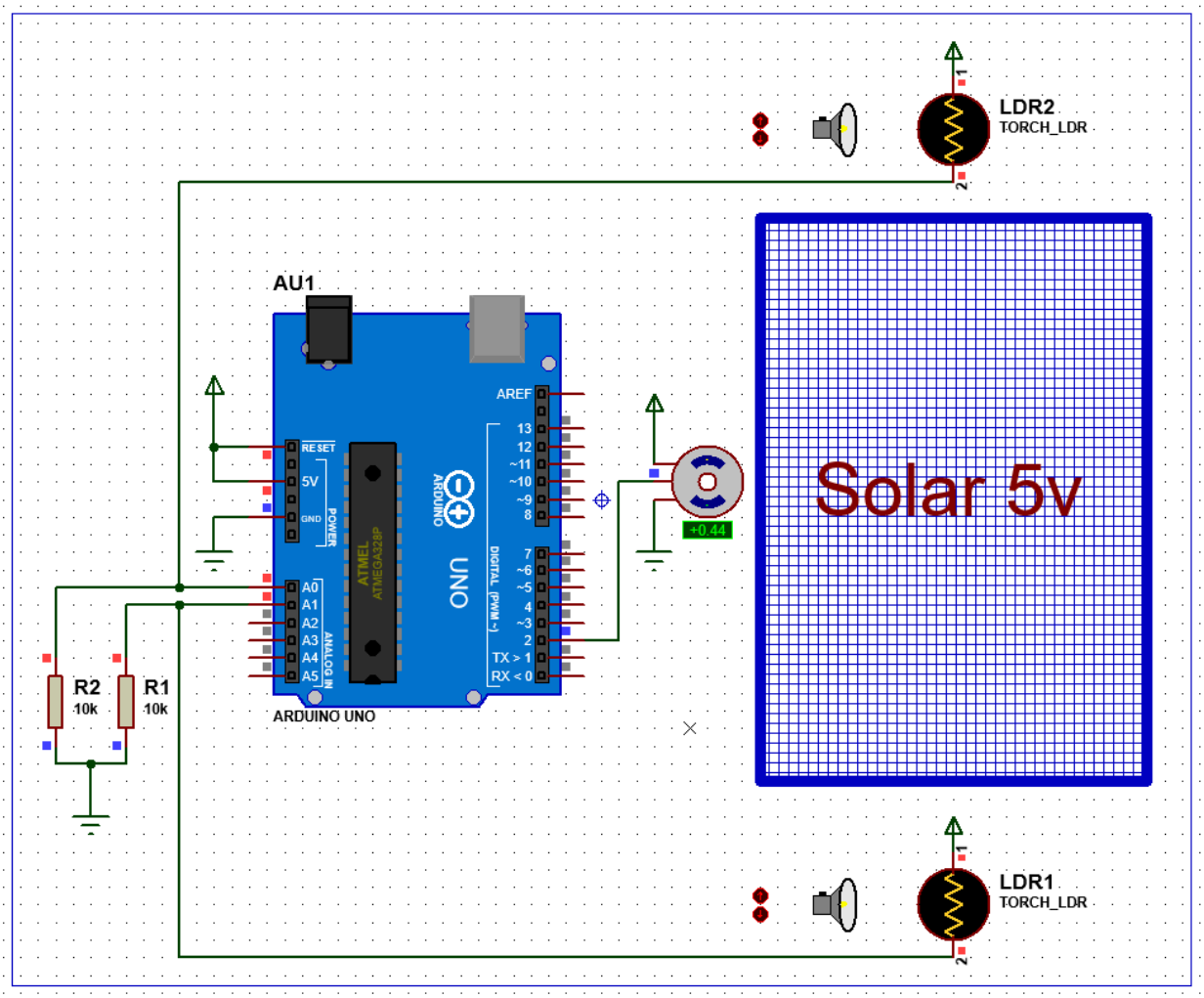

Fig 2: "Circuit Diagram of Sun tracking Solar System".

Procedure:

1. At first, we cut the wood panel into a 60x60 cm frame. Then make some custom stand for horizonal movement of solar.
2. Place the stand in the frame with glue gun.
3. Soldering the solar panel connection in series for getting 5 Volt 3-Watt power supply.
4. Connect the solar panel with LiPo Battery Charger Module Mini TP4056 IC & Connect the module with 3.7V Battery Cell.
5. Attach a Power On/Off Switch with the 3.7V Battery Cell and putting the connection on USB DC 1~5 V to DC 5V Voltage Step Up Boost Module.

Fig 3: "Battery connection with TP4056 IC

6. From the Step-Up boost module, we connect the positive wire into Arduino 5V port and Negative/Ground wire connect into GND port of Arduino.
7. Attach two LDR Sensor into Solar Panel Horizontal side. From LDR sensor, connect 100k resistors to LDR's & attach to GND. The other pin connects to 5V. Now, LDR Sensor is working with servo motor whether light is moves in different direction.
8. Meanwhile attaching LDR sensor, upload the code in Arduino Uno R3. It's full functioning sun tracking solar system.

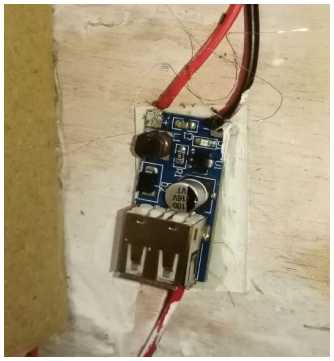

Fig 4: "USB DC 1~5 V to DC 5V Voltage Step Up Boost Module"

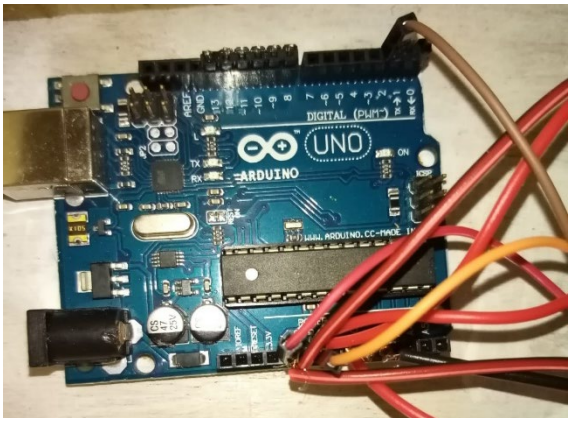

Fig 5: "Arduino Uno Connection".

Fig 6: "Solar Panel & LDR Connection".

Source Code:

```
#include <Servo.h> //includes the servo library
Servo myservo;
#define ldr1 A0 // set ldr 1 Analog input pin of East ldr as an integer
#define ldr2 A1 // set ldr 2 Analog input pin of West ldr as an integer
int pos = 90; // initial position of the Horizontal movement controlling servo
motor
int tolerance = 20; // allowable tolerance setting - so solar servo motor isn't
constantly in motion
void setup(){
myservo.attach(2); // attaches the servo on digital pin 2 to the horizontal
movement servo motor
pinMode(ldr1, INPUT); //set East ldr pin as an input
pinMode(ldr2, INPUT); //set West ldr pin as an input
myservo.write(pos); // write the starting position of the horizontal movement
servo motor
delay(1000); // 1 second delay to allow the solar panel to move to its staring
position before commencing solar tracking
}
void loop(){
int val1 = analogRead(ldr1); // read the value of ldr 1
int val2 = analogRead(ldr2); // read the value of ldr 2
if((abs(val1 - val2) <= tolerance) || (abs(val2 - val1) <= tolerance)) {
//no servo motor horizontal movement will take place if the ldr value is within
the allowable tolerance
}else {
if(val1 > val2) // if ldr1 senses more light than ldr2
{
pos = pos+1; // decrement the 90-degree position of the horizontal servo motor -
this will move the panel position Eastward
}
if(val1 < val2) // if ldr2 senses more light than ldr1
{
pos = pos-1; // increment the 90-degree position of the horizontal motor - this
will move the panel position Westward
 }
}
if(pos > 180) {pos = 180;} // reset the horizontal position of the motor to 180
if it tries to move past this point
if(pos < 0) {pos = 0;} // reset the horizontal position of the motor to 0 if it
tries to move past this point
myservo.write(pos); // write the starting position to the horizontal motor
delay(50);
}
```

Output:

Fig 7: "The Arduino Based Sun Tracking Solar System Basic Structure".

Conclusion:

Finally, the Arduino Uno-based sun tracking solar system provides a unique and effective approach for maximizing solar energy harvesting. This technology maximizes energy output throughout the day by dynamically aligning solar panels with the position of the sun. Its use of Arduino technology demonstrates the promise of accessible and cost-effective automation in renewable energy systems. As we progress towards more sustainable energy solutions, this initiative emphasizes the need to leverage innovative technology to build a greener and more efficient future.